

Monitoring effects of remediation on natural sediment recovery in Sydney Harbour, Nova Scotia

Tony R. Walker · Devin MacAskill ·
Theresa Rushton · Andrew Thalheimer ·
Peter Weaver

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Abstract Chemical contaminants were assessed in Sydney Harbour, Nova Scotia during pre-remediation (baseline) and 3 years of remediation of a former coking and steel facility after nearly a century of operation and historical pollution into the Sydney Tar Ponds (STP). Concentrations of polycyclic aromatic hydrocarbons (PAHs), polychlorinated biphenyls, metals, and inorganic parameters measured in sediments and total suspended solids in seawater indicate that the overall spatial distribution pattern of historical contaminants remains unchanged, although at much lower concentrations than previously reported due to natural sediment recovery, despite remediation activities. Measured sediment deposition rates in bottom-moored traps during baseline were low ($0.4\text{--}0.8\text{ cm year}^{-1}$), but during dredging operations required for construction of new port facilities in the inner Sydney

Harbour, sedimentation rates were equivalent to $26\text{--}128\text{ cm year}^{-1}$. Measurements of sediment chemical contaminants confirmed that natural recovery rates of Sydney Harbour sediments were in broad agreement with predicted concentrations, or in some cases, lower than originally predicted despite remediation activities at the STP site. Overall, most measured contaminants in sediments showed little temporal variability (4 years), except for the detection of significant increases in total PAH concentrations during the onset of remediation monitoring compared to baseline. This slight increase represents only a short-term interruption in the overall natural recovery of sediments in Sydney Harbour, which were enhanced due to the positive impacts of large-scale dredging of less contaminated outer harbor sediments which were discharged into a confined disposal area located in the inner harbor.

T. R. Walker (✉) · T. Rushton · A. Thalheimer
Dillon Consulting Limited,
137 Chain Lake Drive,
Halifax, NS B3S 1B3, Canada
e-mail: tonyrobertwalker@gmail.com

D. MacAskill
Dillon Consulting Limited,
275 Charlotte Street,
Sydney, NS B1P 1C6, Canada

P. Weaver
Sydney Tar Ponds Agency,
1 Inglis Street,
Sydney, NS B1P 6J7, Canada

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Introduction

Sydney Harbour, Nova Scotia has long been subject to effluent and atmospheric inputs of contaminants, including metals, polycyclic aromatic hydrocarbons (PAHs), and polychlorinated biphenyls (PCBs), from a large coking and steel plant which operated

in Sydney from 1901 until it closed in 1988 (Lambert and Lane 2004). Contaminants comprised of coal tar residues were discharged from coking ovens into Coke Ovens Brook, which in turn were discharged into the Sydney Tar Ponds (STP) (Furinsky 2002). The STP forms a small tidal tributary, Muggah Creek, located in the South Arm of Sydney Harbour. Thousands of tons of metal and PCB- (derived from electrical equipment used in the steel plant) and PAH-contaminated sludge were released into Muggah Creek via an aquatic pathway into the receiving marine environment of Sydney Harbour (Lambert et al. 2006). These particle-reactive contaminants bind to fine-grained organic-rich harbor sediments and become the main pathway of exposure and endpoint for contaminants in the marine environment, particularly for benthic organisms (Chapman 1989; Stewart et al. 2002; Leipe et al. 2005).

Numerous studies have linked these effluent and atmospheric contaminants to local ecological and human health impacts in the area (e.g., O'Leary and Covell 2002; Tay et al. 2003). Within Sydney Harbour, the highest chemical contaminant concentrations in sediments were historically reported near Muggah Creek, with concentrations decreasing towards the outer harbor (e.g., Stewart et al. 2001; Lee et al. 2002). Additionally, elevated chemical contaminants previously detected in the digestive glands of American lobster (*Homarus americanus*) resulted in the closure of the commercial lobster fishery in the inner harbor in 1982 (Hilderbrand 1982; King et al. 1993).

Since the closure of coking operations, there have been several attempts to remediate this former industrial site (Campbell 2002). Finally, in May 2004, the governments of Canada and Nova Scotia committed to remediating the STP to reduce potential ecological and human health risks to the environment. A federal and provincial Environmental Impact Statement (EIS) and public Joint Review Panel concluded that remediation would be “unlikely to cause significant negative environmental impacts with the implementation of appropriate and stringent mitigation” (AMEC 2005). A follow-up monitoring study was established to determine the effectiveness of mitigative measures on potential adverse environmental effects during remediation and to verify environmental effect predictions made in the EIS in accordance with the *Canadian Environmental Assessment Act*. Monitoring was designed to assess

changes (both “positive” and “negative”) in the surrounding ecosystem (including groundwater, aquatic, and marine), potentially attributed to remedial activities (Dillon Consulting Limited 2010). This study in Sydney Harbour comprised a subcomponent of the larger monitoring study.

Remediation at the STP site began in 2009 and included in situ remediation using solidification and stabilization (S/S) by mixing contaminated tar pond sediments with Portland cement, transforming it into a durable, solid, low-hydraulic conductivity matrix to reduce contaminant leaching rates into the surrounding environment (ITRC 2011). Although the S/S technology is relatively new, several temporal studies of S/S performance at former industrial sites revealed that environmental conditions have improved to within acceptable limits for many contaminants (Paria and Yuet 2006; ITRC 2011).

The pollution legacy of contaminants in Sydney Harbour sediments is well documented (e.g., Lee et al. 2002) and reached their maxima between 1960 and 1980 coinciding with peak production of coking operations (Barlow and May 2000), but a study by Smith et al. (2009) used radionuclide tracers to predict the time required for natural recovery (or “capping”) of historical contaminants. Our study included ground truth monitoring of current contaminant levels in Sydney Harbour sediments to assess monitored natural recovery (MNR) and to confirm modeled predictions of natural sediment recovery rates reported by Smith et al. (2009). The technique of MNR can have profound economic, human, and ecological health benefits by avoiding sediment removal (or “capping”) activities (Magar and Wenning 2006; Wenning et al. 2006), and benthic organisms respond positively by bioaccumulating less contaminants in their tissues (Leipe et al. 2005).

The general aim of this study was to address potential impacts of remedial activities on surface sediments (currently undergoing natural sediment burial), during baseline and remediation activities at the STP. Specifically, the objectives were (1) to monitor potential disturbance artifacts (either positive or negative) of remedial activities by assessing results of sediment and water quality indicators in Sydney Harbour and (2) to assess contaminant concentrations using MNR techniques to confirm earlier modeled predictions of natural sediment recovery reported by Smith et al. (2009).

Materials and methods

Monitoring stations

Sydney Harbour, Nova Scotia has modest freshwater inputs from Sydney River resulting in estuarine conditions with tidal ranges of 0.9–1.4 m (Gregory et al. 1993; Petrie et al. 2001). Four “areas of assessment” and 11 marine monitoring stations (9–11 m deep) were sampled by boat to monitor variables throughout this study during 2009 baseline (pre-remediation) and 3 years of remediation (i.e., year 1, 2010; year 2, 2011; year 3, 2012) (Fig. 1): area 1—near-field (three stations), area 2—mid-field (three stations), area 3—far-field (two stations), and area 4—Sydney River Estuary (one station). Geographic positions were recorded using hand-held and boat Garmin GPS units.

Water sampling and analysis

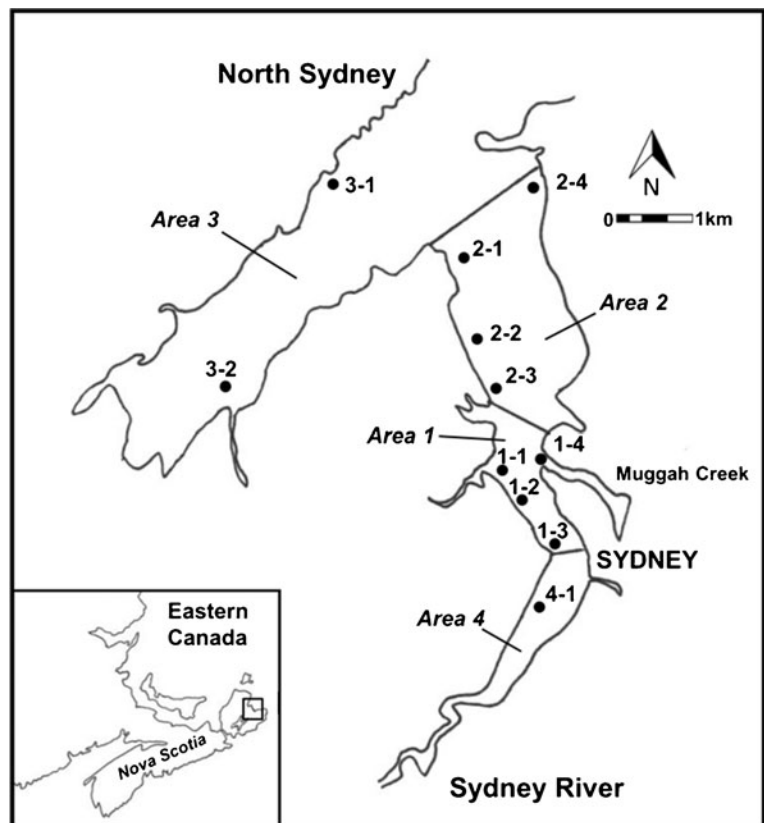
Baseline marine water quality samples for total suspended solids (TSS) were collected monthly between April and August 2009 using a hand-held 2-L

water sampler (Hoskin Scientific®). Remediation phase water samples were collected monthly from September through October each year, except for a 3-month lapse during winter ice cover. Monthly three-point composite samples were collected 1 m below surface (shallow) and 1 m from the bottom (deep) at each station. Composite sampling improves spatial and temporal coverage of stations and reduces contaminant variability (Correll 2001). Care was taken during sampling to prevent collection of surface detritus and bottom sediments. Samples were preserved and analyzed using U.S. Environmental Protection Agency analytical methods based on 160.2/2540-D (APHA 1998; US-EPA 2005). Samples were analyzed by Maxxam Analytics Inc. (Maxxam), a Canadian Analytical Laboratory Association (CALA) certified laboratory.

Sedimentation rates

Bottom-moored sediment traps were deployed for 3 months to measure sediment deposition rates during baseline between April 21 and July 21, 2009 (91 days)

Fig. 1 Marine monitoring stations and assessment areas in Sydney Harbour



and again prior to dredging activities in Sydney Harbour between August 31 and December 9, 2011 (100 days). Sediment traps consisted of PVC pipes (15 cm×122 cm) secured to plastic milk crates bolted to concrete patio stones. Trap dimensions conformed to ratios recommended by Bloesch (1994) for minimizing potential sediment losses due to resuspension (see Nedwell et al. 1993; Walker 2005a; Walker et al. 2008a, b). Depth of sediment accumulation was measured to determine sediment deposition following trap retrieval. Sedimentation rates from stations within the same assessment area were pooled and presented as means for each area ($\pm$ SE, $n=3$).

Sediment sampling and analysis

Sediment samples were collected using an Ekman grab (15×15 cm) (Wildco®). Care was taken to allow surface seawater in the grab to drain away to minimize disturbance of surface sediment before subsampling (Walker and Grant 2009), to retain highly flocculent surface “fluff” which reflect the most recent changes in sediment contaminant concentrations (Milligan and Loring 1997). Subsamples were collected from the top 0–1-cm horizon to capture the most recently deposited sediment material in consideration of low deposition rates in Sydney Harbour (Lee et al. 2002). Three grab samples were collected to create representative composite samples at each station during baseline and remediation (Correll 2001), although triplicate sediment samples were collected in 2011 (year 2) to provide a better understanding of potential variation in chemical concentrations at each site.

Sediment samples were analyzed for grain size, total organic carbon (TOC), PAHs (comprising 18 individual PAH compounds), PCBs, metals (As, Cd, Cu, Pb, Hg, Zn), porosity, sulfides, ammonia, acid volatile sulfides, and simultaneously extracted metals (AVS/SEM ratios). Sediments were sampled during baseline (2009) and 3 years of remediation (2010, 2011, and 2012).

Sediment grain size was analyzed by sieve and pipette based on MSAMS-1978 method. TOC was determined by treating an aliquot of dried sediment sample with sufficient hydrochloric acid (HCl) (1:1) to remove inorganic carbon prior to analysis on dried sediments at 105 °C using a LECO CR-412 Carbon Analyzer (Schumacher 2002). Analysis for low level PAH and aroclor PCB concentrations was based on US-EPA 8270C and US-EPA 8082, respectively.

Metals were analyzed from sediment samples ashed in a muffle furnace at 450 °C overnight; a subsample (0.5 g) of ash was digested in 10 mL of ultrapure concentrated HNO₃. The digest residue (~1 mL) was diluted and filtered through Whatman No. 42 ashless filters and volume increased to 50 mL using ultrapure deionized water. Following acid digestion, metals were analyzed using multi-element inductively coupled plasma-mass spectrometry (ICP-MS) based on US-EPA 6020A (US-EPA 2005).

Quality control

New powder-free nitrile gloves were worn during each sample collection to minimize potential cross contamination. Sediment and seawater samples were transported to the laboratory within 6 h in coolers on ice. Blind field duplicates were collected for every ten samples. Analytical test methods were based on US-EPA (2005), unless otherwise stated.

Data analysis

Excel and Minitab were used to perform standard statistical analyses. Wilcoxon matched-pair signed ranks tests were used to determine significant differences between baseline and remediation unless otherwise stated. In our sampling design, baseline sampling data represents “before” and were compared with subsequent annual monitoring events during remediation, representing “after” treatment. US-EPA (2006) was used to calculate 95 % upper confidence limits (UCL95) for TSS concentrations based on combined shallow and deep water data for each assessment area (including TSS data with below detection limit observations). Concentrations of PAHs and PCBs were compared to the National Oceanographic and Atmospheric Administration criteria for effects range-low (ER-L), corresponding to background concentrations below which the presence of contaminants has little chronic or acute effect on benthic organisms; and effects range-medium (ER-M), above which organisms are very likely to be negatively affected by the presence of contaminants (Jones et al. 1997). Metal and PCB concentrations were compared to Canadian Council of Ministers of the Environment (CCME) Canadian Water Quality Guidelines for the Protection of Aquatic Life probable effect levels (PEL) and the interim sediment quality guidelines (ISQG) (CCME 2007).

Results and discussion

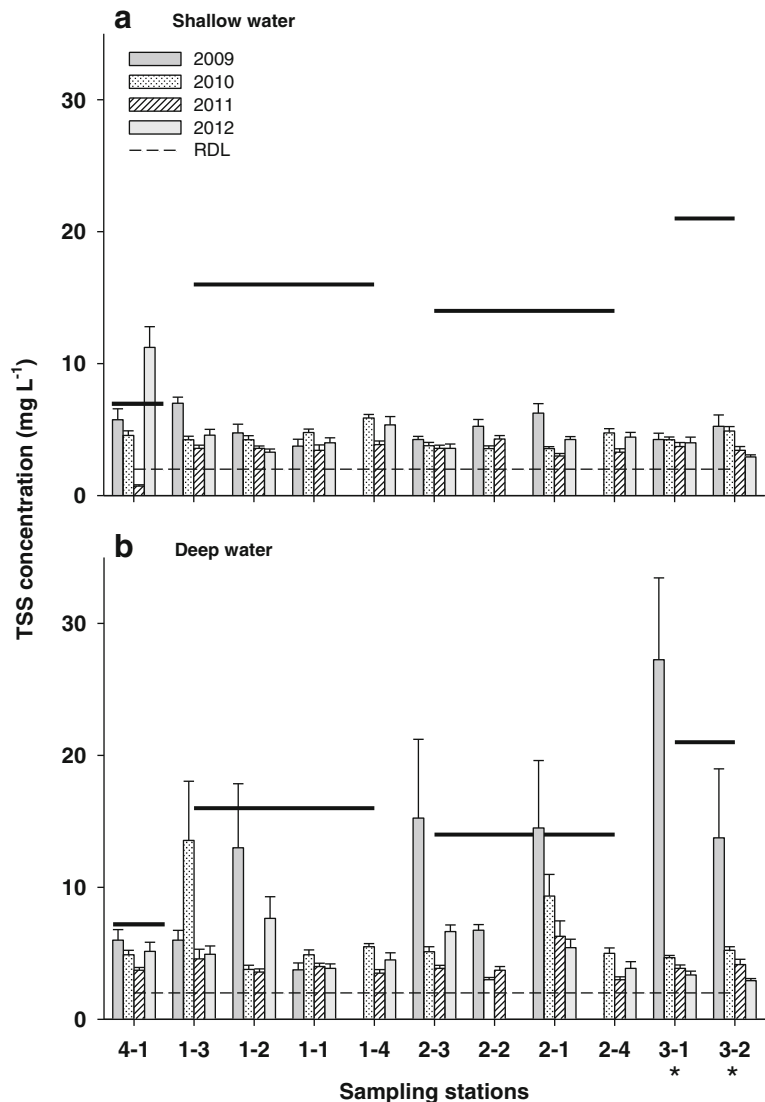
Water quality indicators

Mean monthly TSS concentrations were generally higher in deep water (3.7 ± 0.4 to 27.3 ± 6.2 mg L^{-1}) compared to shallow water stations (3.0 ± 0.1 to 7.0 ± 0.5 mg L^{-1}) (Fig. 2). Higher TSS concentrations at depth may be related to sampling equipment disturbance artifacts or normally higher concentrations encountered at the sediment/water interface from tidal or current resuspension (Sanford et al. 2001). Variations in surface and deep water TSS concentrations were significantly different between monitoring years,

with consistently lower TSS concentrations during remediation activities. A Wilcoxon signed rank test of shallow TSS concentrations during baseline versus year 1, with nine matched observations, yields $z=2.073$, $P=0.0191$ (one-tailed) and baseline versus year 2— $z=2.666$, $P=0.0038$. Mean TSS concentrations in deep water samples were also significantly different between baseline and remediation (year 1, $z=1.836$, $P=0.0332$; year 2, $z=2.547$, $P=0.0054$). Overall, TSS concentrations across all stations decreased in both shallow and deep samples since remediation began (Fig. 3).

Mean TSS concentrations in surface samples did not exceed UCL95s spatially or temporally, whereas

Fig. 2 Mean monthly concentration of TSS in seawater (2009 baseline, *dark gray*; 2010 year 1 remediation, *dotted*; 2011 year 2 remediation, *diagonal lines*; 2012 year 3 remediation, *light gray*). Plotted values are means $\pm$ SE ($n=4$ baseline, $n=9$ year 1, $n=7$ years 2 and 3) from shallow water (**a**) (1 m below surface) and deeper zone (**b**) (1 m from sea bottom). *Solid horizontal lines* indicate UCL95 values for each assessment area. *Dashed lines* indicate detection limits (DL)=2 mg L^{-1} . Far-field stations are indicated (*asterisk*)



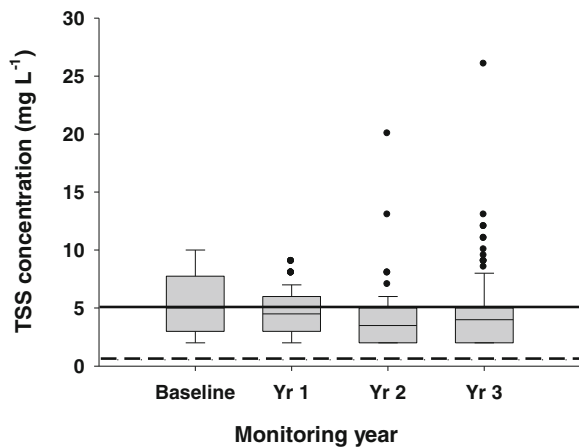


Fig. 3 Box plots of TSS concentrations in seawater represent pooled shallow and deep marine water data. Solid horizontal line indicates area 1 95UCL. Dashed horizontal line indicates DL

deep samples exceeded UCL95s in areas 1, 2, and 3. Furthermore, excessive releases of TSS from the STP site to the marine environment were not detected above measured baseline conditions, suggesting that onsite environmental mitigation measures during remediation activities were implemented. This was also corroborated by estimating mass fluxes of TSS in surface water from the STP site via Muggah Creek, which showed a dramatic decrease between year 1 ($750,000 \text{ kg year}^{-1}$) and year 2 ($225,000 \text{ kg year}^{-1}$) remediation (Dillon Consulting Limited 2012).

An environmental assessment for dredging required for construction of a new marine container terminal in Sydney Harbour reported TSS concentrations of 10 mg L^{-1} (surface) and $50\text{--}100 \text{ mg L}^{-1}$ (bottom), which were higher than those measured during this study (Jacques Whitford 2009). Their estimated sphere of influence for elevated TSS concentrations from dredging was predicted to be highly localized. This was confirmed in water quality data collected during dredging (October 2011 to January 2012), with the highest TSS concentration (73 mg L^{-1}) measured closest to sediment discharge areas (station 3–2). While the Jacques Whitford (2009) model accurately predicted localized elevated TSS in the water column, their predicted re-deposition plume model estimated that concentrations would be $>400 \text{ mg L}^{-1}$, considerably higher than those measured in this study.

Typically, TSS concentrations can vary and depend largely on tides, wind, storm, and rainfall events (from terrigenous run-off), in coastal and estuarine systems due to periodic remobilization of surface sediments

(Calmano et al. 1993; Lintern et al. 2005). Sydney Harbour is also a busy working port handling cruise ships, coast guard vessels, and coal and fuel cargo, which may potentially contribute to temporary sediment resuspension through propeller wash and could account for some of this variability (Ailstock et al. 2004).

Sedimentation rates

Measured sediment deposition rates in bottom-moored traps during baseline were low in all areas ($0.4\text{--}0.8 \text{ cm year}^{-1}$) (Fig. 4). The greatest deposition occurred near Muggah Creek and Sydney River (areas 1 and 4). According to Petrie et al. (2001), flocculation is the most important deposition mechanism in these areas and these low sedimentation rates were comparable to previous estimates made in Sydney Harbour ($0.2\text{--}2 \text{ cm year}^{-1}$) (Stewart et al. 2001; Lee et al. 2002; Smith et al. 2009). Deposition rates were also similar to those reported for Halifax Harbour ($0.1\text{--}1.0 \text{ cm year}^{-1}$) by Buckley et al. (1995) and Cranston (1999), although sedimentation rates measured during dredging operations in Sydney Harbour were equivalent to $26\text{--}128 \text{ cm year}^{-1}$ at stations 2–2 and 2–3 representing huge quantities of sediment for “capping.”

Sediment quality indicators

Generally, sediments contained predominantly silt/clay mixtures with $>70 \%$ for stations located in areas 1 and 4 (Fig. 5). Grain size analysis indicated a decrease in fines (silt/clay) seaward as the marine

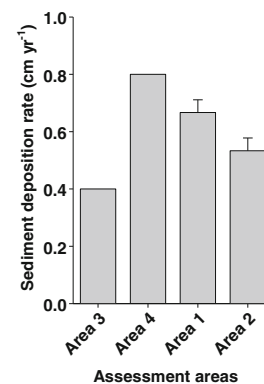
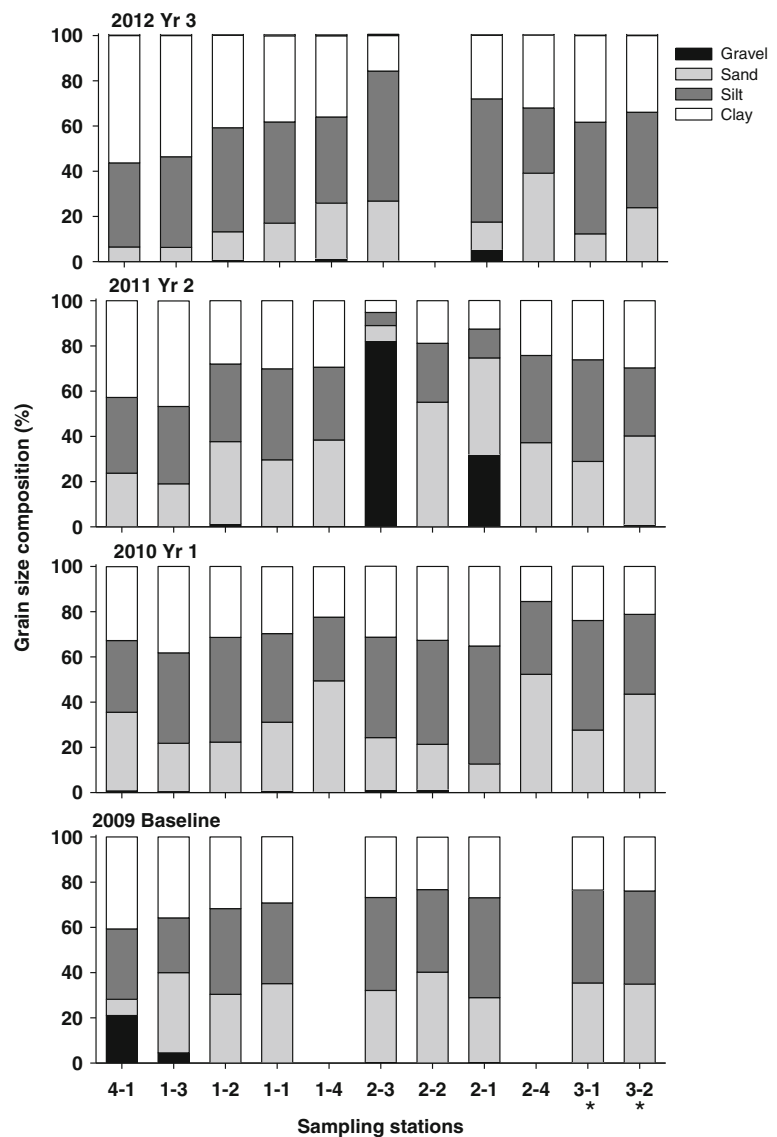


Fig. 4 Mean sediment deposition rates for each assessment area calculated from pooled sediment traps from nine stations. Plotted values in areas 1 and 2 represent means $\pm$ SE ($n=3$)

Fig. 5 Temporal and spatial variations in sediment grain size in surface sediments during baseline and 3 years of remediation



environment becomes more energetic with sandy sediments prevailing in area 3 stations. At most stations during baseline and year 1 remediation, there was little interannual variation in the physical composition of sediment in terms of grain size, except for a few minor changes, likely representing subtle shifts in sampling location due to GPS accuracy (± 10 m), rather than actual changes in grain size. However, during year 2 remediation, there were substantial changes in grain size characteristics in area 2 ranging from gravel to boulder with less abundance of finer sediments (<30 % silt/clay content). During previous sampling (e.g., baseline and year 1), these stations were characterized by 60–80 % silt/clay substrate, suggesting an

uncovering event between sampling events. Changes in sediment grain size at stations 2–1, 2–2, and 2–3 provide empirical evidence that the Sydney Harbour marine environment is a dynamic one in which storm events during winter and fall, in addition to high wave energy in this area, could cause major uncovering events (Lintern et al. 2005). Grain size characteristics measured in year 3 were similar to baseline.

Ammonia, AVS/SEM ratios, redox, sulfide, and porosity sediment characteristics in Table 1 show that stations 4–1 and 1–3 exhibit the highest porosities and concentrations of ammonia and sulfide compared to other stations, presumably reflecting influence of organic inputs from sewage effluent near Muggah

Table 1 Sediment characteristics: ammonia, AVS/SEM ratios, redox, sulfide and porosity at monitoring stations in Sydney Harbour

Station	Year	Ammonia-N ($\mu\text{g g}^{-1}$)	AVS/SEM ratio	Redox potential, Eh (mV)	Sulfide (μM)	Porosity (%)
1-1	<i>Baseline (2009)</i>	<i>30</i>	<i>3.0</i>	<i>41</i>	<i>45</i>	<i>62</i>
	Year 1 (2010)	107	0.2	−87	307	69
	Year 2 (2011)	4	0.4	156	314	69
	Year 3 (2012)	4	0.4	43	134	68
1-2	<i>2009</i>	<i><25</i>	<i>0.5</i>	<i>54</i>	<i>281</i>	<i>66</i>
	2010	29	0.3	−26	239	68
	2011	1	1.3	154	162	68
	2012	2	1.5	51	47	67
1-3	<i>2009</i>	<i>145</i>	<i>ND</i>	<i>34</i>	<i>1</i>	<i>75</i>
	2010	80	0.0	−84	2,000	72
	2011	22	0.3	130	330	79
	2012	8	0.1	10	681	79
1-4	<i>2009</i>	<i>ND</i>	<i>ND</i>	<i>ND</i>	<i>ND</i>	<i>ND</i>
	2010	<25	0.1	−100	1,440	74
	2011	25	0.6	126	193	72
	2012	6	0.2	88	266	71
2-1	<i>2009</i>	<i><25</i>	<i>8.3</i>	<i>190</i>	<i>11.1</i>	<i>58</i>
	2010	<25	22.7	15	2.5	51
	2011	1	0.5	83	60.0	49
	2012	1	1.4	50	27	51
2-2	<i>2009</i>	<i>34</i>	<i>12.1</i>	<i>57</i>	<i>90</i>	<i>65</i>
	2010	<25	0.5	196	118	57
	2011	2	5.0	111	6.2	44
	2012	ND	ND	ND	ND	ND
2-3	<i>2009</i>	<i><25</i>	<i>ND</i>	<i>59</i>	<i>0.9</i>	<i>62</i>
	2010	<25	37.8	25	1.6	56
	2011	3	1.0	100	177	61
	2012	1	0.9	73	32	52
2-4	<i>2009</i>	<i>ND</i>	<i>ND</i>	<i>ND</i>	<i>ND</i>	<i>ND</i>
	2010	<25	ND	−98	<0.3	56
	2011	2	53.8	181	1.1	54
	2012	2	1.4	68	32.4	53
3-1	<i>2009</i>	<i><25</i>	<i>ND</i>	<i>207</i>	<i>0.4</i>	<i>62</i>
	2010	<25	0.9	−37	52.0	60
	2011	10	5.8	111	9.0	67
	2012	5	1.0	55	35	68
3-2	<i>2009</i>	<i><25</i>	<i>ND</i>	<i>85</i>	<i>0.4</i>	<i>61</i>
	2010	<25	0.2	−74	253	66
	2011	15	0.2	77	244	72
	2012	5	0.1	79	365	69
4-1	<i>2009</i>	<i>155</i>	<i>0.1</i>	<i><5</i>	<i>1,140</i>	<i>81</i>
	2010	53	0.1	−100	1,080	76
	2011	17	3.1	102	16	79
	2012	201	0.1	18	376	80

Baseline—italics, remediation phases—upright

ND not determined

Creek, coal dust, and Sydney River. Sulfide concentrations were highly variable during baseline and remediation, with high values in area 1, and at reference station 3–2. Area 1 stations had the lowest AVS/SEM ratios which serve as predictors of toxic effects of organic enrichment and anoxic conditions in sediments (Magar and Wenning 2006). Ratios >1 have excess sulfide available to form insoluble metal sulfides which are relatively nonbioavailable for uptake by benthic organisms, thereby reducing potential toxicity; however, metals can become potentially toxic and bioavailable to benthic organisms with ratios <1 (McGrath et al. 2002).

Although some of the lowest AVS/SEM ratios (<1) were measured at stations closest to Muggah Creek and elsewhere within area 1, there were also low ratios detected in the far-field reference area, so interpretation of contaminant bioavailability may not be entirely related to sulfide availability alone. However, stations in area 2 had AVS/SEM ratios >1 , suggesting that metal sulfide contaminants would be relatively nonbioavailable to benthic organisms (Walker 2005b). Redox potential was highly variable both spatially and temporally, but did not indicate impacts related to remediation activities as they were within baseline ranges.

Likewise, there was some interannual variation in TOC concentrations, with year 1 showing significantly higher TOC concentrations than baseline (Wilcoxon signed rank test, nine matched observations, $z=2.310$, $P=0.0104$, one-tailed); in contrast to baseline versus year 2 which showed no significant differences ($z=1.066$, $P=0.1432$) (Fig. 6). Stations 2–1, 2–2, and 2–3 exhibited coarser material during 2011 (year 2) and also showed a marked decrease in TOC concentrations compared with year 1, as evidenced by the significantly lower TOC concentrations in year 2 versus year 1 ($z=2.547$, $P=0.0054$). Except for stations in area 2, results from sediments collected in the inner harbor are consistent with earlier studies (e.g., Stewart et al. 2001; Lee et al. 2002), suggesting that comparison of sediment chemistry between years is unaffected by temporal changes in substrate type.

Sediment metal, PAH, and PCB quality indicators

Generally, stations near Muggah Creek and Sydney River exhibited higher concentrations of measured parameters, although these declined with increasing

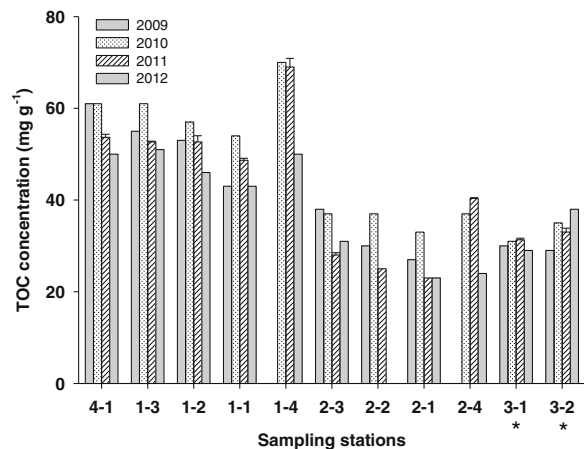


Fig. 6 TOC concentrations in surface sediments (2009, dark gray; 2010, dotted; 2011, diagonal lines; 2012, light gray). Plotted values represent a single composite sample from three grab samples per station (2009, 2010, and 2012) and means from triplicate samples $\pm$ SE (2011). DLs ranged between 0.2 and 0.8 mg g^{-1} depending on grain size composition. Far-field stations are indicated (asterisk)

distance from Muggah Creek. Correspondingly, minimum concentrations were generally found in samples from areas 2 and 3, with lower concentrations measured at far-field stations (3–1 and 3–2). In terms of the spatial distribution of historical contaminants, results were consistent with those of previous studies (e.g., Lee et al. 2002; Smith et al. 2009); however, the maximum concentrations for many of the same contaminants were much lower in our study.

Year 2 error bars ($\pm$ SE) indicate a high degree of precision in surface sediment sampling. There does not appear to be fine or very fine patchiness, respectively, across horizontal (meter) or vertical (millimeter) spatial scales, in contaminant concentrations. Implementation of triplicate sampling in year 2 tended to achieve about the same concentrations as composite sampling during baseline and remediation, providing confidence that reflect harbor conditions at the time of sampling. Therefore, the results were suitable for interannual comparison.

Sampling over 4 years (baseline and 3 years of remediation) showed only minor variation in sediment metal concentrations, suggesting that recent inputs of sediment metals, as a result of remediation activities at the STP, were not detected above measured baseline conditions (Figs. 7 and 8). Where significant differences did occur, concentrations were lower than those measured during baseline, except for minor increases in As and Cd using a Wilcoxon signed rank test, nine

matched observations (As, $z=2.369$, $P=0.0089$; Cd, $z=2.023$, $P=0.0216$, one-tailed) during year 1 compared to baseline. In contrast, there were no significant differences in As and Cd between baseline versus year 2 (As, $z=0.338$, $P=0.3677$; Cd, $z=1.014$, $P=0.1553$). Concentrations of As and Cd were below CCME PEL guidelines and were close to background concentrations

reported for coastal sediments in Nova Scotia (Loring et al. 1996). Arsenic is naturally elevated in rock, soil, and sediments across most of Nova Scotia (Meunier et al. 2010), which may explain why mean As concentrations from all stations each year remained similar (18.5 to $22.2 \mu\text{g g}^{-1}$). Both Pb and Zn concentrations did not vary significantly during year 1 compared to baseline

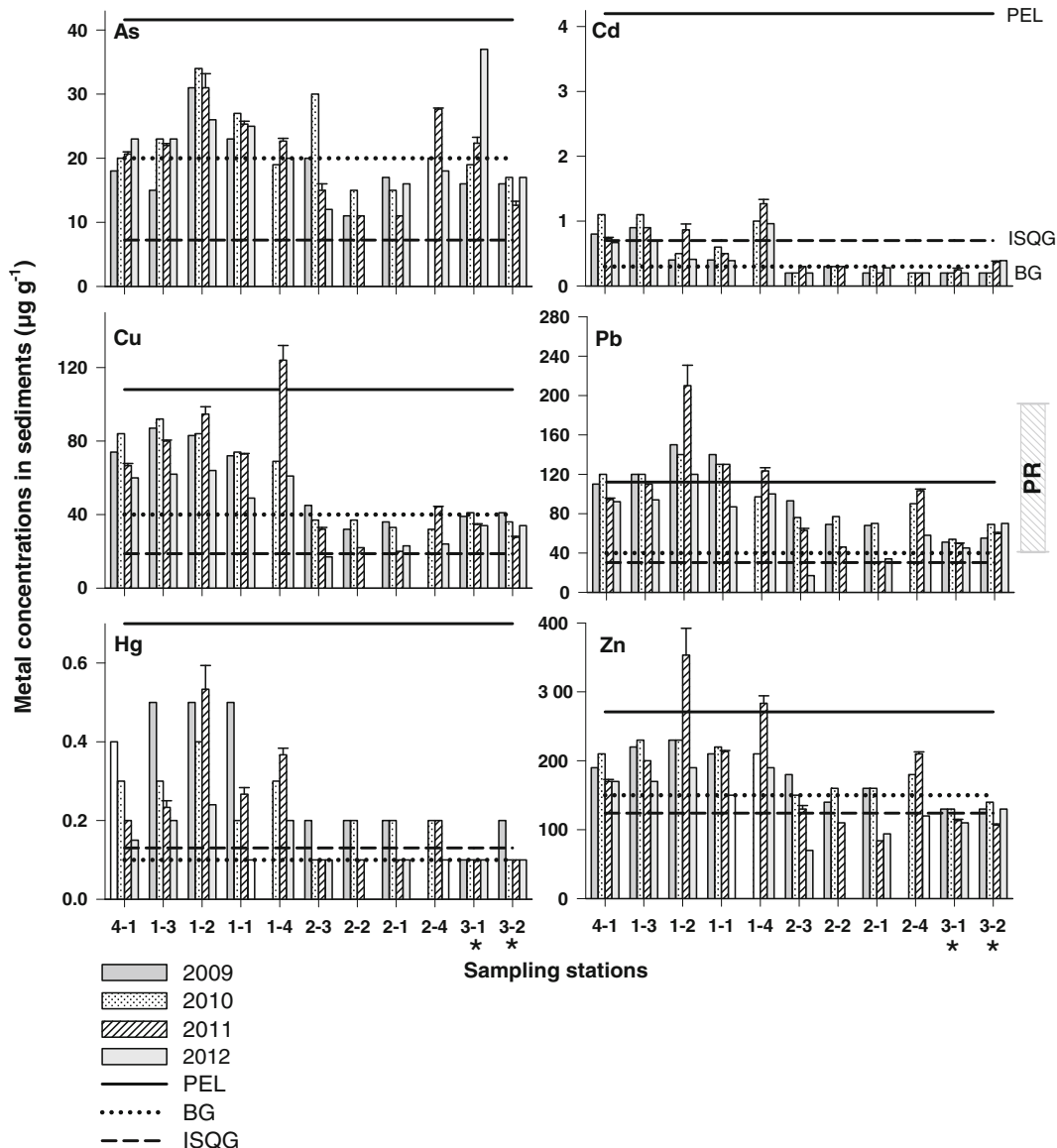
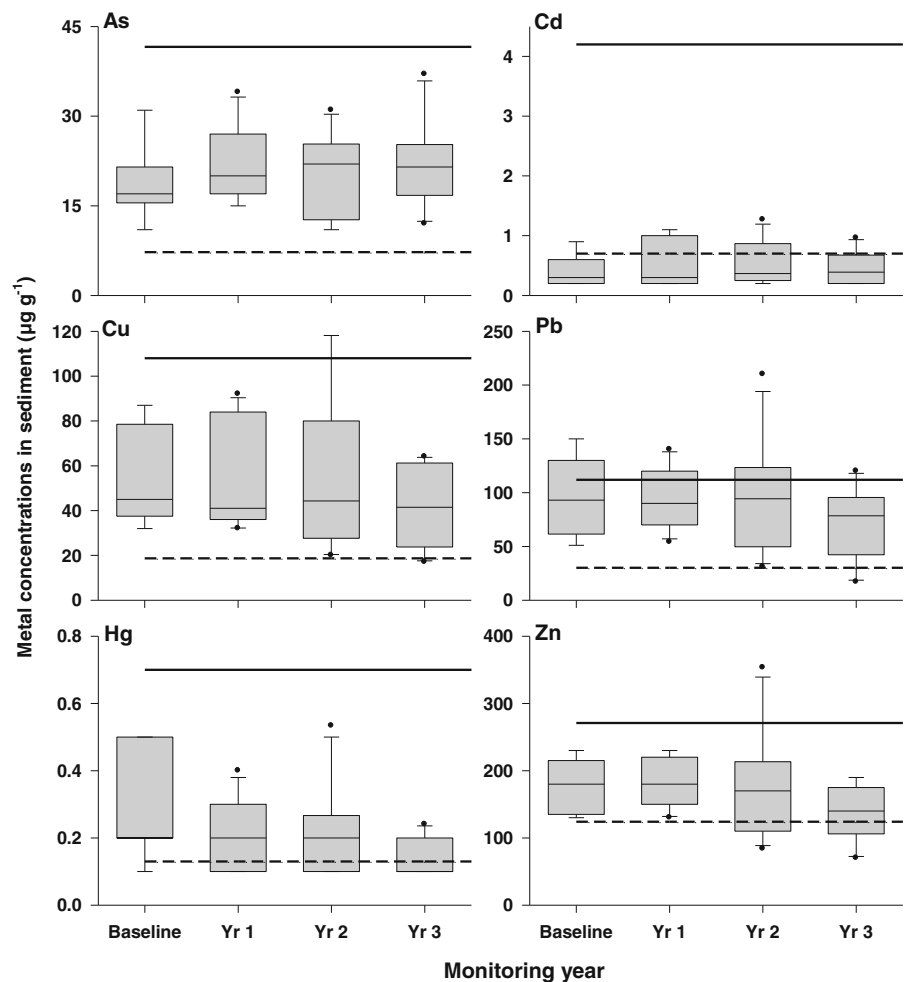


Fig. 7 Temporal and spatial variation in metal concentrations in surface sediments (2009, gray bars; 2010, dotted; 2011, diagonal lines; 2012, light gray) measured by ICP-MS (DLs for As=1; Cd=0.2; Cu=10; Hg=0.1; Pb=1; Zn=50 $\mu\text{g g}^{-1}$). Plotted values represent a single composite sample from three grabs per station (2009, 2010, and 2012) and means from triplicate samples $\pm\text{SE}$

(2011). Solid horizontal lines indicate PEL and dashed lines indicate ISQG values (CCME 2007). Dotted lines indicate background concentrations in coastal sediments for Nova Scotia (Loring et al. 1996). Far-field stations (asterisk) and predicted ranges (PR) for Pb concentrations reported by Smith et al. (2009) are indicated

Fig. 8 Temporal variation of annual metal concentrations in surface sediments



(Pb, $z=0.0$, $P=0.5$; Zn, $z=0.943$, $P=0.1727$) or during year 2 versus baseline (Pb, $z=1.362$, $P=0.0865$; Zn, $z=1.481$, $P=0.0693$).

A small decrease in Cu and Hg concentrations was observed (representing a minor, but positive impact of remediation), suggesting continued natural attenuation of these metals. Concentrations of Cu measured during year 1 was not significantly different from baseline ($z=0.533$, $P=0.2970$), but decreased significantly during year 2 ($z=1.836$, $P=0.0332$). Mercury concentrations decreased significantly during remediation compared to baseline (e.g., year 1, $z=2.201$, $P=0.0139$; year 2, $z=2.380$, $P=0.0086$) and continued to decrease during year 3 (Figs. 7 and 8).

Almost all sediment metal concentrations were below CCME PEL concentrations across all stations, except for Pb at stations close to Muggah Creek, Cu at station 1–4 in 2011, and Zn at stations 1–2 and 1–4

during 2011. The PEL concentration for Pb ($112 \mu\text{g g}^{-1}$) was exceeded at stations 4–1, 1–1, 1–2, and 1–3. Most stations exceeded CCME ISQG concentrations for other metals. Only Cd concentrations were below ISQGs ($0.7 \mu\text{g g}^{-1}$) at all but stations 4–1, 1–3, and 1–2 (Fig. 7).

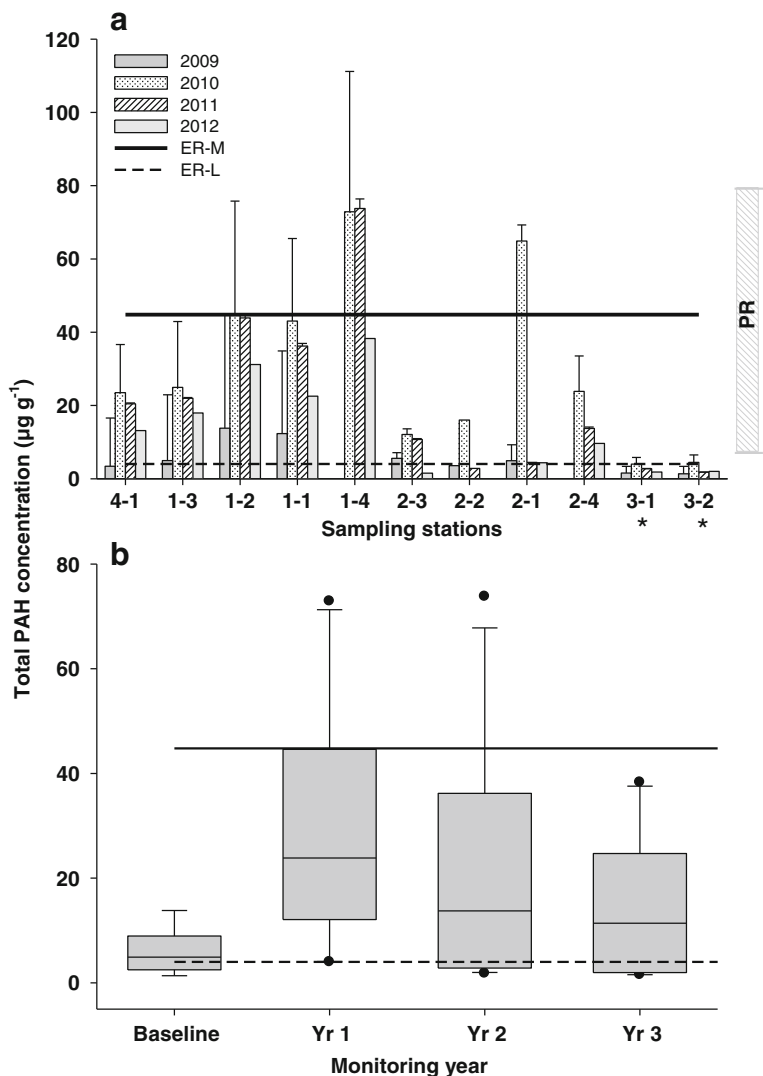
Despite these results, metal concentrations in Sydney Harbour are lower than other industrial harbors in North America (e.g., Tay et al. 2003), and several were below background concentrations reported for coastal sediments in Nova Scotia (Loring et al. 1996). Historical inputs of metal contaminants were likely derived from atmospheric sources from coal combustion when the coking and steel facility was operating (J. Smith, personal communication, March 17, 2011), which is also supported by the sediment metal concentration data which tend to decrease less sharply from Muggah Creek. For example, elevated Pb and Zn concentrations in area 1 stations were consistent with Pb and Zn

signals found in sediment and soil samples close to coal combustion emission sources in Russia (Walker et al. 2003; Walker 2005c).

Temporal variations in total PAH concentrations were significantly different, with generally higher concentrations during remediation compared to 2009 baseline (1.4–13.8 $\mu\text{g g}^{-1}$) (Fig. 9). Wilcoxon signed rank tests revealed that PAH concentrations increased significantly each year compared to baseline (year 1, $z=2.666$, $P=0.0038$; year 2, $z=2.192$, $P=0.0142$). Spatial variations in total PAH concentrations decrease sharply with increasing distance from Muggah Creek, with the lowest concentrations ($<5 \mu\text{g g}^{-1}$) at both far-field stations. The ER-L and ER-M for PAHs (4.02

and $44.8 \mu\text{g g}^{-1}$, respectively; Jones et al. 1997) correspond to the dashed and solid horizontal lines (Fig. 9a, b). Smith et al. (2009) predicted that by 2020 surface concentrations of total PAHs would still exceed ER-M near Muggah Creek, but our monitoring results reveal that most stations were lower than ER-M (except station 2–1). However, total PAH concentrations at most stations still exceed ER-L values. CCME PEL and ISQG guidelines for the protection of marine aquatic life do not exist for total PAHs, but stipulate guidelines for individual PAH compounds instead. However, some individual PAH compounds exceeded ISQGs (all stations) and PELs (some stations).

Fig. 9 **a** Spatial variation of total PAH concentrations in surface sediments (2009, gray bars; 2010, dotted; 2011, diagonal lines; 2012, light gray) measured by GC/MS (DL for individual PAHs= $0.005 \mu\text{g g}^{-1}$). Plotted values represent a single composite sample from three grabs per station (2009, 2010, and 2012) and means from triplicate samples $\pm$ SE (2011). Solid horizontal lines indicate ER-M ($44.8 \mu\text{g g}^{-1}$) and dashed lines indicate ER-L ($4.02 \mu\text{g g}^{-1}$). Far-field stations (asterisk) and predicted ranges (PR) for PAH concentrations reported by Smith et al. (2009) are indicated. **b** Temporal variation of annual total PAH concentrations



Overall, ranges of total PAH concentrations in surface sediments measured in Sydney Harbour between 2009 and 2012 ($1.4\text{--}73.8\text{ }\mu\text{g g}^{-1}$) were at least an order of magnitude lower than ranges determined by Smith et al. (2009) from geochronologies of sediment cores representing concentrations from the 1960s to the 1980s ($200\text{--}500\text{ }\mu\text{g g}^{-1}$). Their study also predicted that total PAH concentrations in surface sediments would eventually decrease to levels between 5 and $80\text{ }\mu\text{g g}^{-1}$ by 2010 (comparable to ranges observed in our study), as a result of natural containment by less contaminated sediments. Another study by Tay et al. (2003) in Sydney Harbour reported total PAH concentrations in sediments collected in 1999 were an order of magnitude higher ($17\text{--}246\text{ }\mu\text{g g}^{-1}$) than those measured in our study. Furthermore, total PAH concentrations in our study were comparable to those in Halifax Harbour ($20\text{--}27\text{ }\mu\text{g g}^{-1}$) (Gearing et al. 1991; Tay et al. 1992). Total PAH concentrations in our reference stations ($1.4\text{--}4.5\text{ }\mu\text{g g}^{-1}$) were also comparable to background levels measured in the Gulf of Maine ($1\text{ }\mu\text{g g}^{-1}$) by Laflamme and Hites (1978).

The increase in total PAH concentrations in sediments during remediation could have occurred because of onsite releases from remedial activities, but the expected signature from a point source such as the STP would have resulted in a more localized increase in concentrations (i.e., centered on area 1 stations) rather than at all stations as evidenced in our results. One possible explanation for the increase in concentrations across all stations (specifically during year 1) could have been caused by resuspension of contaminated sediments as a result of major storms. For example, there were five major storms and hurricanes between 2009 and 2010 (Dillon Consulting Limited 2012). Also, results during baseline could have been “unusually” low due to burial from less contaminated sediments during major storm activity. However, absolute total PAH concentrations during year 1 (2010) were low compared to historical levels. Subsequent measurements of PAHs in 2011 and 2012 revealed that overall concentrations were lower than 2010, but still significantly different from baseline concentrations, although the observed changes in grain size at area 2 stations support the possibility of storm-induced resuspension.

In order to help explain the increase in total PAH concentrations in surface sediments across all stations during remediation, the flux of PAHs leaving the STP

site via Muggah Creek was estimated (Dillon Consulting Limited 2012). The total efflux was estimated as the product of total water outflow multiplied by the average PAH concentration in surface water measured at the mouth of Muggah Creek ($1.1\text{ }\mu\text{g L}^{-1}$). Approximately 100 kg year^{-1} of PAHs was estimated to have left Muggah Creek during year 1 and only 20 kg year^{-1} during year 2. These estimated effluxes are considerably lower than the 777 kg year^{-1} estimated in 2001 by JDAC (2002). Analysis by Lee et al. (2002) at that time also indicated that the concentration of PAHs in the harbor was, in fact, decreasing. Therefore, an increase of PAH concentrations in harbor sediment stations based on a flux $20\text{--}100\text{ kg year}^{-1}$ would not be expected. Further monitoring of total PAH concentrations and fluxes relative to distance from Muggah Creek may lead to a better understanding of the actual transport mechanism.

A further complicating factor in the variation in total PAH concentration may be the presence of coal dust observed in some sediment, particularly those close to the bulk coal unloading wharf at Whitney Pier, located 400 m from the mouth of Muggah Creek. Sediments sampled in 2011 near the coal loading facility revealed PAH concentrations were approximately double the maximum values measured in this study (i.e., $137.8\text{ }\mu\text{g g}^{-1}$) (Dillon Consulting Limited 2012). Coal dust (or “black carbon”) contamination in marine sediments may be responsible for high PAH and TOC concentrations, but may not result in pollution impacts (i.e., negative biological effects) (Chapman et al. 2009; Yan et al. 2009). Black carbon (e.g., soot, coal, coke) has a high binding affinity for PAHs and even small amounts in bottom or suspended sediments can dramatically reduce bioavailability and toxicity of PAHs. For example, Paine et al. (1996) reported that PAHs associated with coal and coke from an aluminum smelter had limited bioavailability and, hence, low toxicity. Although the specific causes for significant increases in PAH concentrations in sediments during remediation compared to baseline remain uncertain, the absolute concentrations are low compared to historical levels. This slight increase represents a short-term interruption in the overall natural recovery of sediments in Sydney Harbour, but the detection of these changes attests to the effectiveness of MNR (Smith et al. 2009; Walker et al. 2013a).

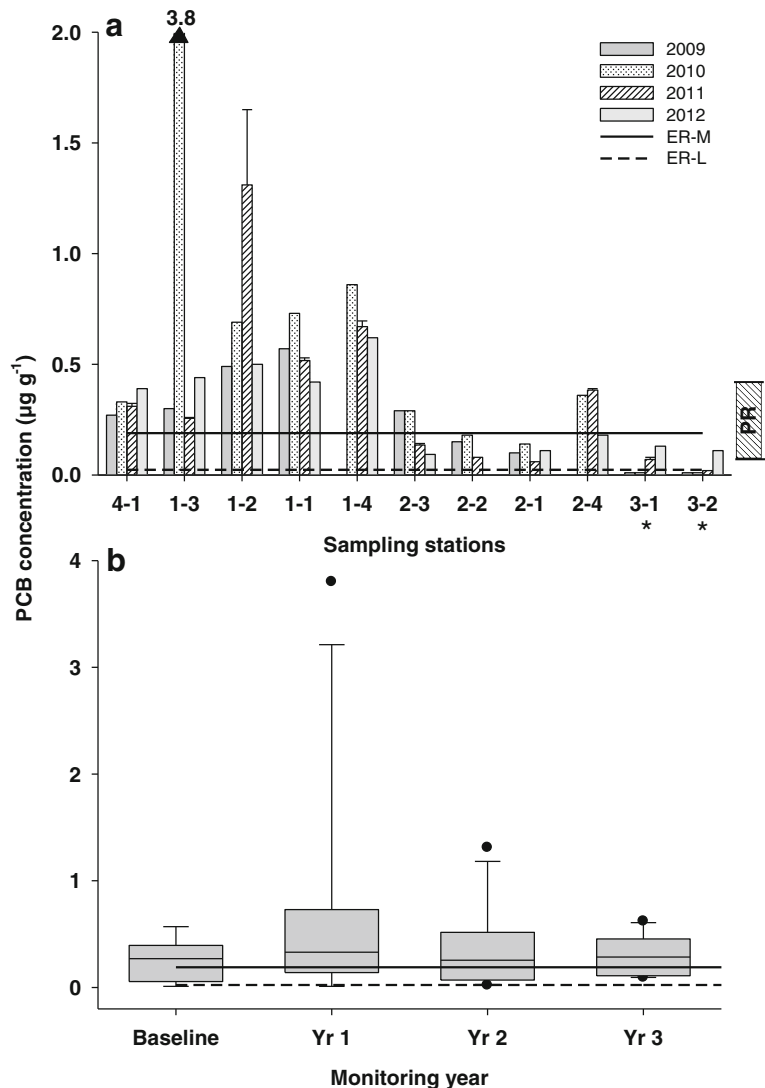
PCB concentrations were significantly higher between baseline and year 1 ($z=2.201$, $P=0.0139$), but

showed no significant differences between baseline and years 2 and 3 ($z=0.533$, $P=0.2970$) (Fig. 10a, b). Stations 4-1, 1-1, 1-2, 1-3, and 2-3 exceeded ER-M for PCBs during baseline and remediation reflecting the historical PCB inputs into the harbor. All stations exceeded ER-L values.

Smith et al. (2009) reported that PCB levels have increased since the 1900s from $<0.005 \mu\text{g g}^{-1}$ in sediments prior to peak activity at the coking facility in the 1960s–1980s up to maximum values of $>1 \mu\text{g g}^{-1}$. PCBs are entirely man-made industrial chemicals (Bowes and Jonkel 1975) and would not have been found in sediments prior to the early 1900s. A study by Tay et al. (2003) reported similar concentrations

with a maximum value of $3.4 \mu\text{g g}^{-1}$ near Muggah Creek comparable to the highest concentration ($3.8 \mu\text{g g}^{-1}$) measured in this study at station 1-3, which is not surprising considering the persistence of PCBs in the environment (Bowes and Jonkel 1975). On the Pacific coast, concentrations of PCBs have been observed in contaminated sediment as high as $8.4 \mu\text{g g}^{-1}$, from industrial sources in British Columbia (Environment Canada 1994), whereas maximum PCB levels of $0.04 \mu\text{g g}^{-1}$ measured in sediment cores from San Francisco Bay (Venkatesan et al. 1999) are more typical where PCB inputs are related to general urbanization rather than to specific industrial inputs.

Fig. 10 **a** Spatial variation of PCB concentration in surface sediments (2009, gray bars; 2010, dotted; 2011, diagonal lines; 2012, light gray) measured by GC/ECD ($\text{DL}=0.01 \mu\text{g g}^{-1}$). Plotted values represent a single composite sample from three grabs per station (2009, 2010, and 2012) and means from triplicate samples $\pm$ SE (2011). Solid horizontal lines indicate ER-M ($0.18 \mu\text{g g}^{-1}$) and dashed lines indicate ER-L ($0.023 \mu\text{g g}^{-1}$) (Jones et al. 1997). PEL and ISQG values were comparable to ER-M and ER-L values and were not plotted (0.189 and $0.0215 \mu\text{g g}^{-1}$, respectively). Far-field stations (asterisk) and predicted ranges (PR) for PCB concentrations reported by Smith et al. (2009) are indicated. **b** Temporal variation of annual PCB concentrations



Assessment of remediation activities and confirmation of modeled predictions of natural sediment recovery

Ongoing recovery of sediments may have been partially enhanced following recent channel and berth dredging activities (October 2011 to January 2012). Dredging was required for a new marine container terminal in Sydney Harbour, which will accommodate large Post-Panamax size ships (Jacques Whitford 2009). While these types of dredging and infilling operations often result in major sediment disturbances, such as resuspension, remobilization, and enhanced bioavailability of historical contaminants (Eggleton and Thomas 2004), it was found that these dredging operations remobilized less contaminated sediment from the outer harbor which increased rates of natural sediment recovery near Muggah Creek (Walker et al. 2013a).

Our sediment trap data collected during dredging indicated enhanced recovery of sediments. For example, measured individual PAH concentrations were at least an order of magnitude lower at station 2–3 following dredging activities compared to pre-dredging (e.g., phenanthrene and pyrene concentrations were 0.81 and 0.98 $\mu\text{g g}^{-1}$ pre-dredging and 0.069 and 0.071 $\mu\text{g g}^{-1}$ post-dredging, respectively).

The primary recovery processes for MNR are natural sediment burial and contaminant transformation to less toxic forms (Magar and Wenning 2006). Several MNR techniques may be used to demonstrate natural recovery processes including measurement of sediment stability, contaminant source control, contaminant burial, contaminant transformation, ecological recovery, and long-term risk reduction (Wenning et al. 2006).

Sediment stability is affected by many factors such as porosity, organic content, grain size, and bioturbation (e.g., Walker et al. 2008b; Walker and Grant 2009). Therefore, quantifying sediment transport and erosion thresholds in coastal systems becomes an important predictive tool in order to understand fluxes of organic matter, particulates, and associated contaminants (Walker and Grant 2009). The apparent attenuation of contaminants and primary recovery processes has been the result of natural sediment burial and contaminant transformation (e.g., microbial degradation in the case of PAH compounds) and weathering to less toxic forms.

Therefore, if natural containment is sufficiently rapid and effective, it may not be necessary to

undertake large investments in capital and environmental resource management to clean harbor sediments, such as the use of more aggressive remedies such as capping, dredging, and the use of amendments. Natural recovery processes are likely to provide the most cost-effective solution for sediments in Sydney Harbour and MNR would support future management decisions (Walker et al. 2013a).

Interestingly, Smith et al. (2009) projected concentration ranges of Pb, PAH, and PCBs for 2010 that were comparable to those measured here and by Walker et al. (2013a). For example, they reported a projected range of concentrations of Pb in surface sediments by 2010 (25–110 $\mu\text{g g}^{-1}$), following natural sediment containment, which compared favorably to the minimum and maximum range of measured Pb concentrations of 31–210 $\mu\text{g g}^{-1}$ in our study (Fig. 7). A similar pattern also emerged with PAH concentrations between predicted (5–80 $\mu\text{g g}^{-1}$) and measured values (1.4–73.8 $\mu\text{g g}^{-1}$ [this study, Fig. 9a]), and similar with PCB concentrations between predicted (0.02–0.4 $\mu\text{g g}^{-1}$) and measured values in this study, except for 3.8 $\mu\text{g g}^{-1}$ measured at station 1–3 in 2010 (Fig. 10a).

As the harbor sediments become less contaminated, they become more hospitable for benthic infauna, as evidenced by our subtidal benthic community indicator monitoring (Dillon Consulting Limited 2010). This resulted in greater abundance and more diverse benthic communities (species richness) in Sydney Harbour, when compared to studies in similar areas in 1999–2000 (Stewart et al. 2002). Increasing concentrations of benthic infauna may result in higher rates of bioturbation and the reworking of oxygen, nutrients, and contaminants from deeper sediments to surface sediments, thereby potentially slowing the rates of natural remediation (Nedwell and Walker 1995; Simpson et al. 1998). Measurement of ecological recovery over time can be much more challenging than measuring sediment recovery over time, because ecological recovery tends to be slower and more difficult to quantify. Increases or decreases in benthic communities may not necessarily indicate changes in surface sediment chemistry (e.g., Stewart et al. 2002). Instead, they may indicate changes in exposure-mediated environmental loadings not related to sediments (Magar and Wenning 2006). In other words, decreases in environmental loadings due to natural recovery may not be immediately reflected in decreased biological concentration trends, if these trends are overshadowed by other influences

such as sewage effluent or reduced oxygen concentration (Stewart et al. 2002).

Assessments of the time-dependent performance of S/S at remediated sites on metals, PCBs, PAHs, and coal tar constituents concluded that former industrial sites improved environmental conditions to within acceptable limits for most contaminants (Paria and Yuet 2006; ITRC 2011). This study presents data in support of the positive impacts (for most parameters) on Sydney Harbour sediments as a result of stringent onsite mitigative measures implemented during S/S remediation activities at the STP site and of natural sediment recovery predictions made by Smith et al. (2009).

While sediment contaminant concentrations are lower than those previously reported, most of the contaminant inventory remains intact below the sediment surface, with the potential risk for future contaminant disturbance or resuspension (Wenning et al. 2006). The greatest risk of potential sediment erosion would be expected during high-energy conditions, including storms, flood events, and human disturbances, especially ship wake from cruise ships, which can be significant in Sydney Harbour.

Despite several regulatory exceedances for sediment contaminants, there were no comparable effects observed in marine organisms. For example, there was a lack of chemical signature in annual tissue assessments of blue mussels (*Mytilus edulis*) and rock crabs (*Cancer irroratus*) (Walker et al. 2013b, c). Contaminants measured in blue mussels were below the reported Canadian Food Inspection Agency levels (Walker et al. 2013c) and rock crab contaminant concentrations were much lower than those reported by Chou et al. (2002) in rock crabs analyzed from stations in the inner Bay of Fundy (Walker et al. 2013b). Other studies have shown similar ecological patterns during periods of natural attenuation of historical contaminated sediments. For example, Leipe et al. (2005) found that PAH and metal-enriched sediments in the Western Baltic became less toxic to the benthic ocean quahog (*Arctica islandica*) if left undisturbed during natural attenuation and decrease in bioavailability of contaminants. Historical contaminant levels that resulted in the closure of the commercial lobster fishery in the inner Sydney Harbour in 1982 (Hilderbrand 1982; Uthe and Musial 1986) now appear to be present at much lower concentrations. For example, PAH concentrations in rock crab hepatopancreas tissue was undetectable ($<0.05 \mu\text{g g}^{-1}$) and are much lower than those reported in lobster by Ernst et al. (1999) which were up to $13 \mu\text{g g}^{-1}$ (Walker et al. 2013b).

Conclusions

The annual monitoring of most marine sediment parameters in Sydney Harbour has not detected evidence of substantial contaminant releases into the marine environment as a result of remediation activities at the STP site. However, total PAH concentrations in sediments increased significantly during remediation compared to baseline monitoring, but absolute concentrations were lower compared to historical levels. This slight increase represents only a short-term interruption in the overall natural recovery of sediments in Sydney Harbour despite the land-based remediation activities at the STP site. These recent ground truthed results of sediment chemistry support the modeled predictions made by Smith et al. (2009) of natural containment of the main inventory of contaminants deposited during the 1960s–1980s by layers of cleaner surficial sediments. The continuing trend of natural sediment containment may have also been enhanced as a result of recent dredging activities in Sydney Harbour. Further monitoring of these ecological indicators becomes increasingly important, whether the aim is to assess impacts of industrial activities (e.g., remediation of a former industrial sites or harbor dredging activities) or to gauge the rates of natural sediment recovery using MNR.

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